Reference Modular and Jacobi Estimates for a Half-Filled Fermion Interval

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Scope

Seven notes on the half-filled reference state of a finite fermion interval and its commuting Jacobi matrix: all-degree Jacobi estimates, exact occupation weights and particle–hole pairing, logarithmic weighted modular control, the sharp logarithmic boundary moment, the sine-kernel endpoint, the fixed-index spectral expansion and the occupation logarithm. Each statement is for the specified reference kernel.

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All-Degree Jacobi Estimates and Conditional Spectral Localization

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Abstract. A degree-uniform alternating-walk estimate is proved for a finite Jacobi family, without restricting polynomial degree relative to matrix size. A positive Toeplitz majorant yields a quantitative spectral-tail estimate. We derive the corresponding abstract layer-cake bound and explain the conditional logarithmic modular rate when additional Wilson-specific Coulomb and reference-eigenmode hypotheses hold. The sufficient finite-part route developed here remains conditional on its displayed moving-window hypotheses. Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy* proves the fixed-positive-scaled-mass finite part by a different route. The separate leading law with bounded remainder is established in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*.

Model and dependency boundaries

Fix $\mu > 0$, $N = 2L$, and $r = \mu/(N+\mu)$. The reference symmetric Jacobi matrix is J_N with off-diagonal bonds $j(N-j)/N$, $1 \le j < N$. Define $T_N = J_N/N$, and let $\Delta_N = S_N + D_N$ be the scalarized difference between the compressed infinite-chain massive and massless Wilson covariances. Only the fixed reference matrices B_N and J_N commute. The covariance perturbation Δ_N is **not** assumed to commute with either reference matrix.

For $p = 2n+1 \ge 3$, the series for the comparison function is

$$\eta(t) = \sum_{n=0}^{\infty} a_n t^{2n+1}, \qquad a_n = \frac{\pi\binom{2n}{n}\binom{4n}{2n}}{(2n+1)16^n}, \qquad a_0 = \pi.$$

The covariance decomposition, positive Wilson majorant, staggered pairing inequality, and Jacobi spectral-tail estimate are established for this Wilson projector in the companion *Exact Spectral Majorants for a Wilson–Dirac Interval*. The all-degree alternating-walk estimate below supplies a separate finite-Jacobi result. The global reference-mode and moving-window estimates invoked later remain additional assumptions.

An all-odd-degree alternating-walk estimate

Theorem (all-degree alternating-walk bound). For every integer $N \ge 2$ and every odd integer $p \ge 1$, let T_N be the symmetric tridiagonal matrix with bond $\beta_j = j(N-j)/N^2$, $j = 1, \dots, N-1$. Then

$$\boxed{\sum_{\substack{1 \le a < N \\ a \text{ odd}}} \left| \sum_{i=0}^{N-a-1} (-1)^i (T_N^p)_{i,i+a} \right| \le \frac{4\sqrt{p}\, 2^{-p}}{N}.} \tag{S1}$$

The estimate applies also when the polynomial degree is at least the matrix dimension, and it does not require a parity restriction on the dimension.

Proof. For a nearest-neighbor path $\omega = (\epsilon_1, \dots, \epsilon_p)$ with steps $\epsilon_k \in \{+1, -1\}$ and positive odd net displacement, write its bond offsets as integers $b_1, \dots, b_p$. Its contribution at starting site i is the zero-extended path weight

$$f_\omega(i) = \prod_{\ell=1}^{p} \beta_+\Big(\frac{i+b_\ell}{N}\Big), \qquad \beta_+(z) = \max\{z(1-z), 0\}.$$

If the common positive support is empty, the path contributes nothing. Otherwise define the scaled real-variable product $P(x) = \prod_{\ell=1}^{p} \beta_+(x+b_\ell/N)$, so that $f_\omega(i) = P(i/N)$. Its positive support is a bounded interval (a, b); P is a polynomial of degree $2p$ on that interval, vanishes at the endpoints, and has all roots real and outside the open support. Let $a_\ell = 4\beta_+(x+b_\ell/N) \in [0,1]$. As $|\beta'(z)| = \sqrt{1-4\beta(z)}$ in $(0,1)$, differentiating the product and applying Cauchy–Schwarz gives

$$\begin{aligned} |P'(x)| &\le 4^{1-p} \sum_\ell \sqrt{1-a_\ell} \prod_{k\neq\ell} a_k \\ &\le 4^{1-p}\sqrt{p}\left[\sum_\ell (1-a_\ell)\prod_{k\neq\ell} a_k^2\right]^{1/2} \le 4^{1-p}\sqrt{p}. \end{aligned} \tag{S2}$$

The last sum is at most one because each $a_k^2 \le a_k$, and the resulting sum is a nonnegative subcollection of terms in $\prod_k (a_k + (1-a_k)) = 1$.

The logarithm of P is strictly concave on its positive support (a sum of logarithms of positive linear distances to the real roots), so P' has at most one zero there. Real-root interlacing, including repeated-root multiplicities, implies P'' has at most two zeros there. Consequently P' has at most three monotonic intervals and interior total variation at most $6 \sup|P'|$. Zero extension across the two endpoints contributes at most two derivative jumps, and therefore

$$\mathrm{TV}_{\mathbb{R}}(P') \le 8 \sup|P'| \le 32\sqrt{p}\, 4^{-p}. \tag{S3}$$

Now let $g(y) = P(y/N)$ on the support and zero elsewhere. Its distributional second derivative is a finite signed measure, including both possible endpoint atoms. Scaling gives

$$\|g''\|_{\mathrm{TV}} \le \frac{32\sqrt{p}\, 4^{-p}}{N}. \tag{S4}$$

For **any** compactly supported continuous function g whose second distributional derivative is a finite measure, the bilateral discrete second difference obeys

$$\Delta^2 g(i) = g(i+1) - 2g(i) + g(i-1) = \int_{\mathbb{R}} (1-|x-i|)_+ \, g''(dx). \tag{S5}$$

The cardinal hat kernels satisfy the exact partition-of-unity identity $\sum_{i\in\mathbb{Z}} (1-|x-i|)_+ = 1$ for every real x. Thus

$$\sum_i |\Delta^2 g(i)| \le \|g''\|_{\mathrm{TV}}.$$

Because $\sum_i (-1)^i \Delta^2 g(i) = -4\sum_i (-1)^i g(i)$, the alternating sum of **one path** is bounded by

$$\left|\sum_i (-1)^i g(i)\right| \le \frac{1}{4}\|g''\|_{\mathrm{TV}} \le \frac{8\sqrt{p}\, 4^{-p}}{N}. \tag{S6}$$

For odd p, exactly 2^{p-1} unrestricted step sequences have positive net displacement; a step sequence with a displacement $a \ge N$ has zero weight. Summing (S6) over this superset of contributing paths proves (S1). No operation depends on whether the power exceeds the matrix dimension. □

Coefficient consequence for the stated covariance decomposition. Here $D_N^{\mathrm{st}} = D_N$ is the transverse term of the scalarized Pauli decomposition in *Exact Spectral Majorants for a Wilson–Dirac Interval*, section 2. In the fixed scalar ordering its same-parity entries vanish; for odd $a > 0$ write

$$(D_N)_{i,i+a} = \tfrac{1}{2}(-1)^i d_a, \qquad d_a = 2(D_N)_{0,a}.$$

These identities follow by Fourier transforming the transverse off-orbital symbol proportional to $a_r(k)e^{\mp ik/2}$ and applying the cell staggering, where $a_r(k) = r\cos(k/2)/\sqrt{\sin^2(k/2) + r^2\cos^2(k/2)}$. In particular

$$|d_a| \le \frac{1}{2\pi}\int_{-\pi}^{\pi} a_r(k)\,dk = d_*(r) := \frac{2r}{\pi\sqrt{1-r^2}}\,\operatorname{arsinh}\frac{\sqrt{1-r^2}}{r}.$$

Let $a_n = \pi\binom{2n}{n}\binom{4n}{2n}/[(2n+1)16^n]$ be the coefficient of t^{2n+1} in $\eta(t)$. The coefficient bound

$$a_n 2^{-(2n+1)} \le [2\sqrt{2}\,n(2n+1)]^{-1}$$

and the displayed staggered Fourier bound show, using (S1),

$$\left|\mathrm{Tr}\{D_N^{\mathrm{st}} N T_N^{2n+1}\}\right| \le 4d_*(r)\sqrt{2n+1}\,2^{-(2n+1)},$$

hence

$$\boxed{\begin{aligned}&\sum_{n\ge 1} a_n \left|\mathrm{Tr}\{D_N^{\mathrm{st}} N T_N^{2n+1}\}\right| \le \zeta(3/2)\,d_*(r),\\ &\sum_{n>K} a_n \left|\mathrm{Tr}\{D_N^{\mathrm{st}} N T_N^{2n+1}\}\right| \le \frac{2d_*(r)}{\sqrt{K}}.\end{aligned}} \tag{S7}$$

The covariance splitting is imported from the spectral-majorant paper; the scalar Fourier bound and its normalization have been made explicit above. Convergence to the nonlinear finite constant uses a separate all-degree argument, proved in Theorem 7 of *Finite Part of Wilson–Dirac Relative Entropy* and sharpened in Theorem 6 of *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals*.

Positive spectral tail: recalled theorem and its algebraic prerequisite

The tail estimate below is a restatement of the critical spectral-tail theorem in *Exact Spectral Majorants for a Wilson–Dirac Interval*, section 3. The scalar derivation is included to identify the inputs to the conditional localization argument; it is not a separate theorem of priority.

Write G_N for the positive Toeplitz section of

$$f_r(q) = \tfrac{1}{2}\left(1 - \frac{|\cos q|}{\sqrt{\cos^2 q + r^2\sin^2 q}}\right).$$

Positivity is immediate from $f_r \ge 0$ and the Fourier rank-one representation. With $u_q = N^{-1/2}(1, e^{iq}, \dots, e^{i(N-1)q})$, independent finite-sum reconstruction gives

$$\|J_N u_q\|^2 = \underbrace{\frac{N^2-1}{3N^2}}_{\beta_N} + \underbrace{\frac{2(N^2-1)(N^2-4)}{15N^2}\cos^2 q}_{c_N^2}.$$

For the reference spectral projector $P_R = 1_{|J_N|>R}$, use $P_R \le J_N^2/R^2$ in the quadratic-form sense and $P_R \le I$. The exact positive nuclear decomposition and the elementary inequality $\min(1, a+b) \le \min(1,a) + b$ give

$$\operatorname{Tr}(G_N P_R) \le N \int f_r(q) \min\left(1, \frac{c_N^2 \cos^2 q}{R^2}\right) \frac{dq}{2\pi} + \frac{\beta_N \operatorname{Tr} G_N}{R^2}.$$

Rationalization of f_r, the substitution $x = |\cos q|$, and the bound

$$H(\rho) = \int_0^1 \sqrt{1-x^2}\, \min\{x^{-2}, \rho^{-2}\}\, dx \le 2/\rho$$

lead to the all-size estimate

$$\boxed{\operatorname{Tr}(G_N P_R) \le \min\left\{\frac{\mu}{\pi}, \sqrt{\frac{2}{15}}\frac{\mu^2}{\pi R} + \frac{\mu}{3\pi R^2}\right\}.} \tag{S8}$$

The $N = 2$ case is handled directly because $c_2 = 0$. Set $\kappa = \sqrt{2/15}$ and $D_\mu = (4\mu/3 + \kappa\mu^2)/\pi$. Then (S8) proves, for **all** $R > 0$,

$$\nu_N((R,\infty)) := \operatorname{Tr}(G_N P_R) \le D_\mu / R, \tag{S9}$$

by using the mass bound for $R \le 1$ and the two rational terms for $R \ge 1$. This is a proof about the **specified positive Toeplitz majorant**. Applying it to the modular correction additionally requires the exact, model-specific inequalities $|< v_j, S_N v_j >| \le < v_j, G_N v_j >$ and $|< v_j, D_N v_j >| \le 2\sqrt{q_j b_j (1-b_j)}$, which follow for this Wilson model from the companion spectral-majorant theorem, not from scalar tail algebra alone.

Uniform finite Coulomb gates and the logarithmic exponent

The finite reference-mode equation and its exact Liouville normal form are

$$(p_N u'_{\sigma,N})' = (\sigma c - \alpha_N p_N) u_{\sigma,N}, \qquad p_N = N \sin(s/N), \quad \alpha_N = (1 - N^{-2})/4, \quad c = \lambda/2,$$

$$\boxed{w''_{\sigma,N} + \left(\frac{1}{4} - \frac{\sigma c}{p_N} + \frac{1}{4p_N^2}\right) w_{\sigma,N} = 0, \qquad w = \sqrt{p_N}\, u.} \tag{S10}$$

The potentially misleading N^{-2} terms cancel *exactly*; the symbolic calculation was performed afresh. For $h = 2 + \lambda$, assume the uniform forbidden-arc estimate $\int_0^{512h} u_+^2 \le h^{103} \exp(2I_N)$, whereas the allowed-arc approximation is controlled by two-sided relative errors. The finite one-sided step has the logical form

$$U_{\sigma,N} \in [1 - \delta_N(h),\ 1 + \delta_N(h) + C_{\text{in}} h^{103}/x], \quad x = \log N,$$

with $\delta_N(h) \le 1/2$. The elementary logarithmic inequality then gives

$$|e_j| \le \log(1 + C_{\text{in}} h^{103}/x) + 3\delta_N(h) + C_\eta h^3/N^2. \tag{S11}$$

Range restriction. The term $C_{\text{in}}h^{103}/x$ need not be small when h approaches a fixed multiple of $\sqrt{\log N}$. One must not replace $\log(1+z)$ by $O(z)$ under an *unstated smallness assumption* across this full window. The one-sided logarithmic bound is essential to the weighted argument.

For the specified matching constant $C_0 = 300000$, let

$$c_w = (2C_0)^{-1/2} = 1/\sqrt{600000}, \qquad T = \tfrac{1}{2}c_w\sqrt{x}.$$

On the positive-arc window $h \le c_w\sqrt{x}$, the matching error obeys the **exact deterministic upper inequality** $\exp(C_0h^2)/N^2 \le \exp(-3x/2)$; the remaining errors $h/\sqrt{N}$ and h^3/N^2 also vanish uniformly as $x \to \infty$. Hence, once model-dependent matching constant C_a and arc identity are accepted, the $\delta \le 1/2$ and geometric gates hold eventually, with a starting size independent of the mass. This is a quantifier statement, not a practically reachable numerical cutoff.

The conventional reference-mode envelope $|e_j| \le 800\log(2+|\lambda_j|)$ is a **separate global hypothesis**. A finite-element eigenvalue computation cannot establish it uniformly in dimension. The published Coulomb-function and Whittaker asymptotic theories explain the appropriate type of turning-point analysis but do not prove (S11) with these Wilson-specific constants: see NIST DLMF §§13.7, 33.12, and 2.8.

Derivation of the $1/103$ modular rate, conditional on the reference identities

Quantified moving-window hypotheses for this derivation. Assume there exist finite constants $C_{\log}, C_{\text{in}} > 0$ and an even N_0, independent of N, j, such that for every even $N \ge N_0$, with $x = \log N$ and $h_j = 2 + |\lambda_j|$,

$$|e_j| \le C_{\log}\log(1 + C_{\text{in}}h_j^{103}/x) \quad \text{whenever } h_j \le c_w\sqrt{x}, \qquad |e_j| \le 800\log h_j \quad \text{for every } j.$$

These are the direct spectral hypotheses used below. The preceding arc discussion motivates sufficient inputs for them; it does not establish them for this model. In particular, the error terms in (S11) must be bounded and absorbed to obtain the displayed hypothesis before (S13) can be invoked.

For any finite positive measure ν with tail $\nu((t,\infty)) \le D/t$, and any $\alpha > 1$, Tonelli and the beta/reflection identity give

$$\boxed{\int_0^\infty \log(1+(t/\rho)^\alpha)\,d\nu(t) \le \frac{\pi D}{\rho\sin(\pi/\alpha)}.} \tag{S12}$$

This constant is sharp within the **abstract positive-tail class**, using truncated Pareto-tail measures; it does not imply sharp Wilson asymptotics.

Divide the Wilson positive spectral weights at $t = 2$ and $t = T = c_w\sqrt{x}/2$. On the middle interval $2 < t \le T$, use $2 + t \le 2t$, (S11), and (S12) with $\alpha = 103$ and $\rho = x^{1/103}/(2C_{\text{in}}^{1/103})$. On the high interval, use the global $800\log(2+t)$ envelope and the tail bound. The resulting explicit **conditional** estimate is

$$\begin{aligned}\operatorname{Tr}(G_N|E_N|) \le{}& \frac{C_{\log}C_{\text{in}}4^{103}Q_\mu}{x} + 2C_{\log}C_{\text{in}}^{1/103}\pi\csc(\pi/103)\,D_\mu x^{-1/103} \\ &+ 800D_\mu\left[\frac{\log(2+T)}{T} + \frac{1}{2}\log(1+2/T)\right],\end{aligned} \tag{S13}$$

where $Q_\mu = \mu/\pi$. Therefore the positive component is $O_\mu(x^{-1/103})$.

For the staggered part, the required occupation input is stated directly: there exist $C_V < \infty$ and an even N_V such that, for every even $N \geq N_V$,

$$\sum_j e_j^2 b_j(1-b_j) \leq C_V/\log N.$$

This is sufficient together with the diagonal pairing estimate. A bound on $\operatorname{Tr}\sqrt{B_N(I-B_N)}$ alone does not imply it. The stronger $O((\log N)/N^2)$ variance estimate is supplied separately by the "Positive weighted modular control" theorem of *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*, section "Global envelope and weighted consequences." With the stated variance input, Cauchy–Schwarz gives

$$\operatorname{Tr}(E_N^2 B_N(I-B_N)) = O(x^{-1}), \quad |\operatorname{Tr}(D_N^{\mathrm{st}} E_N)| = O(\sqrt{\mu}\, x^{-1/2}). \tag{S14}$$

The transverse conclusion in (S14) uses precisely the displayed variance input and the model-specific pairing. The longitudinal conclusion (S13) separately requires the uniform reference-mode estimates in (S11), with errors absorbed into its stated moving-window logarithmic bound, and the global modular-error envelope. Slepian's discrete-prolate theorem identifies the common finite-dimensional eigenvectors of B_N and J_N and their spectral ordering; it does not establish these Wilson-specific quantitative estimates. Nor do the general special-function references establish them. For $C_0 = 300000$, the proof's chosen sufficient gate $T \geq 1024$ demands $\log N \geq 2,516,582,400,000$. The formula is a mathematically asymptotic bound, **not a practically sized error certificate**.

The assembled modular correction thus has the conditional rate

$$\mathcal{F}_L = \operatorname{Tr}(\Delta_N E_N) = O_\mu((\log L)^{-1/103}). \tag{S15}$$

Here the implied constants are allowed to depend on **fixed** μ, while the abstract reference-mode constants and sufficiently large starting size for the modal envelope need not.

Finite-part bounds and order of limits

The elliptic calculation uses the two-bond formula with **elliptic parameter** $z = 1 - r^2$, which differs from the DLMF elliptic modulus $k = \sqrt{z}$ (complementary modulus $k' = r$). It checks the expansion

$$\mathcal{P}_L = \frac{\mu^2}{6}\log L + \mu^2\left(\frac{1}{6}\log\frac{8}{\mu} - \frac{1}{4}\right) + O_\mu\left(\frac{\log L}{L}\right).$$

The nonlinear scalar limit remains

$$c_\eta = \sum_{n=1}^{\infty} \frac{1}{2n(4n+1)(4n+2)(4n+3)} = \frac{11}{18} - \frac{\pi}{12} - \frac{\log 2}{2} > 0,$$

with analytic tail $0 < c_\eta - \sum_{n=1}^{K} \ldots \leq 1/(384K^3)$. Therefore the finite-part constant simplifies **exactly** to

$$\boxed{C(\mu) = \mu^2\left(\frac{13}{36} - \frac{\pi}{12} - \frac{\log\mu}{6}\right).}$$

These scalar calculations do not establish compact-uniformity of the complete relative-entropy remainder as $\mu \to 0$. The conditional assembly below proves its $1/103$ conclusion only under its displayed modular hypotheses. The fixed-μ finite part also has a direct proof in Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy*, with the stronger remainder $O_\mu(1/\log L)$ and without those moving-window hypotheses.

The conditional assembly has the form

$$D_L(\mu) = \mathcal{P}_L + \mathcal{Q}_N + \mathcal{F}_L - \Delta S_L$$

with rates $O_\mu(\log L/L)$, $O_\mu((\log L/L)^{1/3})$, $O_\mu((\log L)^{-1/103})$, and the entropy-difference estimate $O_\mu(1/\log L)$. When **all** separately stated assumptions hold with compatible quantifiers, the modular rate dominates and yields the claimed $1/103$ expansion. Assembling bounds is not the same as independently proving every antecedent.

Scope of the conditional result

The all-degree alternating-walk theorem is proved for the specified tridiagonal matrices. The positive spectral tail is proved for the specified Toeplitz majorant and is identified with the Wilson covariance in the companion spectral-majorant theorem. The inverse-logarithmic Wilson modular rate derived here additionally requires global reference-mode, occupation-number and Coulomb turning-point estimates. Neither a finite numerical sample nor a general special-function reference establishes these model-specific hypotheses. The $1/103$ implication established here remains conditional on its stated modal estimates. For the same fixed-positive-scaled-mass Wilson comparison, the finite part and an $O_\mu(1/\log L)$ remainder follow from Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy*. This does not verify the moving-window or Coulomb-matching hypotheses of the present sufficient route.

Mathematical references

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4. D. Slepian, "Prolate Spheroidal Wave Functions, Fourier Analysis, and Uncertainty V: The Discrete Case," *Bell System Technical Journal* **57** (1978), 1371–1430, especially equations (14), (21), and (129)–(131), doi:10.1002/j.1538-7305.1978.tb02104.x (finite common eigenvectors and ordering only).

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Exact Reference Occupation Weights and Particle-Hole Pairing

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Abstract

We calculate the total occupation-fluctuation weight of the finite half-filled sine-kernel reference covariance, retaining all finite-size terms, and identify exact particle–hole symmetries relevant to the Wilson–Dirac relative-entropy problem. An explicit harmonic-number/trigamma identity gives the logarithmic growth and its first algebraic corrections. The chiral involution pairs the eigenvalues of the commuting reference Jacobi matrix and the reference covariance and makes the trace of a Wilson positive majorant against every odd reference function vanish. The formula evaluates the half-filled free-fermion number fluctuation with explicit finite-size corrections; the exact symmetries constrain, but do not evaluate, the Wilson longitudinal modular residual.

1. Exact finite reference geometry

For even `N=2L`, consider the real symmetric matrix

$$(B_N)_{ij} = \begin{cases} \frac{1}{2}, & i = j, \\ \dfrac{\sin(\pi(i-j)/2)}{\pi(i-j)}, & i \neq j, \end{cases} \qquad 0 \le i, j < N.$$

The associated irreducible Jacobi matrix has `J_(j-1,j)=J_(j,j-1)=j(N-j)/N`. They commute. Let $\Gamma_N = \mathrm{diag}(1, -1, 1, -1, \dots, -1)$. Since the off-diagonal entries of B_N vanish at even nonzero separations, while the nonzero off-diagonal entries connect opposite parities,

$$\boxed{\Gamma_N B_N \Gamma_N = I - B_N, \qquad \Gamma_N J_N \Gamma_N = -J_N.} \tag{1}$$

Since J_N has simple spectrum, each common eigenpair (λ, b) of (J_N, B_N) has an exactly paired eigenpair $(-\lambda, 1-b)$. In particular, for every odd function f defined on the spectrum of J_N, $\Gamma_N f(J_N)\Gamma_N = -f(J_N)$.

Let G_N denote the positive Wilson majorant obtained by compression of the scalar two-orbital Fourier multiplier $g_r(k)I_2/2$, after the interval-preserving unitary that produces B_N. Its only nonzero matrix entries connect indices of the same parity; therefore

$$\boxed{\Gamma_N G_N \Gamma_N = G_N.} \tag{2}$$

For paired Jacobi eigenvectors $v, \Gamma_N v$, their majorant weights are equal:

$$\langle \Gamma_N v, G_N \Gamma_N v \rangle = \langle v, G_N v \rangle.$$

Consequently, $\mathrm{Tr}[G_N f(J_N)] = 0$ for every odd f. In particular, define the *exact* reference modular discrepancy

$$E_N = \log(I - B_N) - \log B_N + N\eta(J_N/N),$$

where η is the odd half-filled reference scaling function. Functional calculus gives $\Gamma_N E_N \Gamma_N = -E_N$, hence the exact cancellation

$$\boxed{\mathrm{Tr}(G_N E_N) = 0.} \tag{3}$$

This does **not** imply $\mathrm{Tr}[(A_N - B_N)E_N] = 0$: the covariance difference is not G_N. Equation (3) supplies a symmetry check and prevents incorrectly interpreting the majorant's signed first moment as the desired modular term.

2. Exact occupation-fluctuation trace

Define

$$V_N := \mathrm{Tr}[B_N(I - B_N)] = \sum_{j=1}^{N} b_j(1 - b_j).$$

Proposition. For every even integer $N \geq 2$,

$$\boxed{V_N = \frac{N}{4} - \frac{2}{\pi^2} \sum_{\substack{1 \leq d < N \\ d \text{ odd}}} \frac{N - d}{d^2}.} \tag{4}$$

Writing $H_n = \sum_{k=1}^{n} 1/k$ and $\psi_1(z) = \sum_{k=0}^{\infty} (z + k)^{-2}$, equivalently,

$$\boxed{V_N = \frac{2}{\pi^2}\Big(H_N - \tfrac{1}{2}H_{N/2}\Big) + \frac{N}{2\pi^2}\psi_1\Big(\frac{N+1}{2}\Big).} \tag{5}$$

As $N \to \infty$ along even integers,

$$\boxed{V_N = \frac{1}{\pi^2}\left(\log N + \gamma + \log 2 + 1 - \frac{1}{6N^2} + \frac{7}{20N^4} + O(N^{-6})\right).} \tag{6}$$

Proof. The diagonal of B_N contributes $N/4$ to $\mathrm{Tr}\, B_N - \mathrm{Tr}\, B_N^2$. At distance d, the two off-diagonals contain $N - d$ entries each. Their squares vanish for even d and equal $1/(\pi^2 d^2)$ for odd d. This proves (4).

The classical odd-integer identity $\sum_{d \geq 1,\ d \text{ odd}} d^{-2} = \pi^2/8$ converts the un-cancelled term to a positive exterior tail:

$$V_N = \frac{2}{\pi^2} \sum_{\substack{1 \leq d < N \\ d \text{ odd}}} \frac{1}{d} + \frac{2N}{\pi^2} \sum_{\substack{d \geq N \\ d \text{ odd}}} \frac{1}{d^2}.$$

The first sum equals $H_N - \frac{1}{2}H_{N/2}$, and the second equals $\frac{1}{4}\psi_1((N+1)/2)$, proving (5). The Euler–Maclaurin expansions of harmonic numbers and ψ_1, at the displayed half-integer argument, yield

$$2\Big(H_N - \tfrac{1}{2}H_{N/2}\Big) = \log N + \gamma + \log 2 + \frac{1}{6N^2} - \frac{7}{60N^4} + O(N^{-6}),$$

$$\frac{N}{2}\psi_1\left(\frac{N+1}{2}\right) = 1 - \frac{1}{3N^2} + \frac{7}{15N^4} + O(N^{-6}).$$

Adding these expansions gives (6). All sums and steps are exact until the final asymptotic expansion. □

3. Consequences for weighted modular estimates

The exact trace fixes the total occupation-fluctuation weight: $V_N \sim \pi^{-2} \log N$. For any $\delta \in (0, \frac{1}{2})$, the number of reference eigenvalues in $[\delta, 1-\delta]$ obeys

$$\#\{j : b_j \in [\delta, 1-\delta]\} \leq \frac{V_N}{\delta(1-\delta)} = O_\delta(\log N).$$

This elementary counting consequence is not an estimate on the reference modular discrepancy. A stronger sufficient condition for finer reference-modular error estimates would be

$$\mathrm{Tr}[E_N^2 B_N (I - B_N)] = \sum_j e_j^2 b_j (1 - b_j) = O((\log N)^{-1}),$$

uniformly in even N, which (4)–(6) alone do not imply. *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* supplies the stronger bound $\mathrm{Tr}[E_N^2 B_N (I - B_N)] = O(\log N / N^2)$, using a compact-interior expansion and a global linear logit envelope. The total occupation weight still grows like $\pi^{-2} \log N$; decay comes from control of the modular errors. The exact particle–hole and majorant cancellations alone do not provide that control.

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Logarithmic Weighted Modular Control for the Half-Filled Sine Kernel

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Abstract

For the finite interval of a half-filled infinite discrete sine-kernel projection, we establish an explicit logarithmic upper bound on the occupation-weighted quadratic discrepancy between its exact modular generator and the established commuting-Jacobi scaling function. The proof uses an elementary trace-class estimate for the half-line sine-kernel Hankel blocks and a scalar logarithmic inequality. Combined with the independently established Wilson–Dirac positive-majorant pairing estimate, this shows that the *transverse* contribution to the fixed-scaled-mass relative-entropy modular correction is sublogarithmic. An elementary fractional spectral-moment bound yields a considerably weaker sufficient condition for the leading logarithmic law than uniform boundedness of the modular-error operator. The separate argument in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* supplies the longitudinal bound and leading entropy law. Neither the present estimate nor that leading law determines the finite part or its proposed inverse-logarithmic remainder rate.

1. Reference operators and established inputs

Let $N = 2L \geq 2$. On $\mathbb{C}^N$ define the half-filled sine-kernel covariance

$$(B_N)_{jk} = \begin{cases} \frac{1}{2}, & j = k, \\ \dfrac{\sin(\pi(j-k)/2)}{\pi(j-k)}, & j \neq k, \end{cases} \qquad 0 \leq j, k < N.$$

Let J_N be the real-symmetric tridiagonal matrix with zero diagonal and $(J_N)_{j-1,j} = j(N-j)/N$ for $1 \leq j < N$. These matrices commute, $0 < B_N < I$, and $\|J_N/N\| \leq 1/2$. The strict occupation bounds follow from the fact that a nonzero finitely supported sequence has Fourier transform that cannot vanish identically on either of the two nonempty half-circle frequency arcs.

Set

$$W_N = B_N(I - B_N), \quad K_N = \log(I - B_N) - \log B_N,$$
$$\eta(t) = \pi t \, {}_3F_2\left(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}; 1, \tfrac{3}{2}; 4t^2\right), \quad E_N = K_N + N\eta(J_N/N).$$

The generalized hypergeometric series has nonnegative coefficients and converges at its argument 1. Therefore

$$A_\eta := \pi \, {}_3F_2\left(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}; 1, \tfrac{3}{2}; 1\right) < \infty, \qquad |N\eta(\lambda/N)| \leq A_\eta|\lambda|$$

for every $|\lambda| \leq N/2$. We use the separately proved reference Jacobi moment estimate

$$M_{2,N} = \mathrm{Tr}(J_N^2 W_N) \leq \frac{32}{\pi^2}(H_N + 1), \quad H_N = \sum_{k=1}^{N} \frac{1}{k}. \tag{1}$$

The sharper identity $M_{2,N} = (3\pi^2)^{-1} \log N + O(1)$ is available, but unnecessary below.

2. Explicit trace-class estimate for the reference crossing

Let P be the orthogonal projection on $\ell^2(\mathbb{Z})$ onto the Fourier arc $(-\pi/2, \pi/2)$, and let χ_N project onto N consecutive sites. Then $B_N = \chi_N P \chi_N$, as an operator on the interval. Define $X_N = \chi_N P(I - \chi_N)$. The full-line projection identity yields

$$X_N X_N^* = B_N(I - B_N) = W_N, \qquad \operatorname{Tr} \sqrt{W_N} = \|X_N\|_1. \tag{2}$$

For either exterior half-line, after reversing the interior index, the cross-boundary matrix has entries

$$\frac{\sin\{\pi(i+j+1)/2\}}{\pi(i+j+1)}, \qquad 0 \le i < N, \quad j \ge 0.$$

Write $H^{(0)}_{ij} = 1/(i+j+1)$. The sine factor is a difference of two complex exponentials; each gives diagonal unitary modulation of $H^{(0)}$. Therefore the nuclear norm of each half-line block is at most $\pi^{-1}\|H^{(0)}\|_1$.

The elementary Laplace representation

$$H^{(0)}_{ij} = \int_0^\infty e^{-t(i+j+1)}\, dt$$

is a rank-one Bochner integral. Its integrand has nuclear norm

$$\mathcal{W}_N(t) = e^{-t} \frac{\sqrt{1 - e^{-2Nt}}}{1 - e^{-2t}}.$$

Consequently, for the union of both exterior half-lines,

$$\operatorname{Tr} \sqrt{W_N} = \|X_N\|_1 \le \frac{2}{\pi} I_N, \qquad I_N = \int_0^\infty \mathcal{W}_N(t)\, dt. \tag{3}$$

For $0 < t \le 1$, the concavity inequality $1 - e^{-2t} \ge (1 - e^{-2})t$ and $1 - e^{-2Nt} \le \min\{2Nt, 1\}$ give

$$\mathcal{W}_N(t) \le \frac{1}{1 - e^{-2}} \min\left\{ \sqrt{\frac{2N}{t}}, \frac{1}{t} \right\}.$$

For $t \ge 1$, $\mathcal{W}_N(t) \le e^{-t}/(1 - e^{-2})$. Splitting the integral at $t = 1/N$ and $t = 1$ gives

$$\boxed{I_N \le \frac{\log N + 2\sqrt{2} + e^{-1}}{1 - e^{-2}}.} \tag{4}$$

All constants are independent of N. This is a deliberately nonoptimal estimate.

3. Uniform quadratic modular estimate

Theorem 1. For every even $N \geq 2$,

$$\boxed{\mathrm{Tr}(W_N E_N^2) \leq \frac{16}{\pi} I_N + 2A_\eta^2 M_{2,N} \leq \frac{16(\log N + 2\sqrt{2} + e^{-1})}{\pi(1-e^{-2})} + \frac{64A_\eta^2}{\pi^2}(H_N + 1).} \tag{5}$$

In particular, $\mathrm{Tr}(W_N E_N^2) = O(\log N)$.

Proof. For $b \in (0,1)$ let $k = \log((1-b)/b)$ and $s = \sqrt{b(1-b)} = 1/(2\cosh(k/2))$. Since $\cosh(k/2) \geq k^2/8$,

$$b(1-b)k^2 = s^2 k^2 \leq 4s. \tag{6}$$

Functional calculus gives $\mathrm{Tr}(W_N K_N^2) \leq 4\,\mathrm{Tr}\sqrt{W_N}$. Because B_N, J_N, K_N, E_N and W_N commute,

$$\mathrm{Tr}(W_N E_N^2) \leq 2\,\mathrm{Tr}(W_N K_N^2) + 2\,\mathrm{Tr}\{W_N[N\eta(J_N/N)]^2\} \leq 8\,\mathrm{Tr}\sqrt{W_N} + 2A_\eta^2 M_{2,N}.$$

Substitute (1), (3), and (4). □

Qualification. The estimate (5) is far weaker than the numerically observed decay of the weighted variance and does not itself prove a decaying variance bound. The stronger estimate $O(\log N/N^2)$ is established in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*. Its value is that it is independent of any unproved pointwise modular-eigenvalue comparison.

4. Consequence for the Wilson–Dirac modular correction

For the mass-scaled Wilson–Dirac covariance perturbation Δ_N, use its established full-line-projector decomposition compressed to the interval, $\Delta_N = S_N + D_N$, with positive operator G_N. In the common eigenbasis v_j of B_N, J_N, put $b_j = \langle v_j, B_N v_j\rangle$, $\lambda_j = \langle v_j, J_N v_j\rangle$, $q_j = \langle v_j, G_N v_j\rangle$, $e_j = \langle v_j, E_N v_j\rangle$. Established inputs are

$$\mathrm{Tr}\, G_N \leq M_\mu := \frac{\mu}{\pi}, \quad |\langle v_j, S_N v_j\rangle| \leq q_j, \quad |\langle v_j, D_N v_j\rangle| \leq 2\sqrt{q_j b_j(1-b_j)}.$$

Corollary 2 (transverse modular correction). For each fixed $\mu > 0$,

$$\boxed{|\mathrm{Tr}(D_N E_N)| \leq 2\sqrt{\frac{\mu}{\pi}\,\mathrm{Tr}(W_N E_N^2)} = O_\mu(\sqrt{\log N}) = o(\log N).} \tag{7}$$

Proof. Apply the pairing bound, sum over common eigenvectors and use Cauchy–Schwarz. Then apply Theorem 1. □

Combining (7) with the exact two-bond asymptotic established in the companion Wilson reference-geometry result, bounded nonlinear correction via the Jacobi fourth moment, and bounded Wilson cross-boundary entropy difference yields the leading-order reduction

$$\boxed{\mathcal{D}_L(\mu) = \frac{\mu^2}{6}\log L + \mathrm{Tr}(S_N E_N) + O_\mu(\sqrt{\log L}), \qquad N = 2L.} \tag{8}$$

The leading coefficient follows if $\mathrm{Tr}(S_N E_N) = o(\log L)$. Neither (7) nor (8) proves this condition alone; *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* supplies the stronger estimate $O_\mu(1)$.

5. Fractional spectral moments and weaker sufficient hypotheses

The established positive-majorant spectral-tail estimate is

$$\nu_N((t,\infty)) := \sum_{|\lambda_j|>t} q_j \le \min\left\{M_\mu, \frac{a_\mu}{t} + \frac{b_\mu}{t^2}\right\}, \quad a_\mu = \sqrt{\frac{2}{15}}\frac{\mu^2}{\pi}, \quad b_\mu = \frac{\mu}{3\pi}. \tag{9}$$

Lemma 3 (uniform fractional majorant moment). For every $0 < \alpha < 1$ and every even N,

$$\boxed{\sum_j q_j |\lambda_j|^\alpha \le M_\mu + \frac{\alpha a_\mu}{1-\alpha} + \frac{\alpha b_\mu}{2-\alpha}.} \tag{10}$$

Proof. Use layer-cake: $\sum q_j|\lambda_j|^\alpha = \alpha \int_0^\infty t^{\alpha-1}\nu_N((t,\infty))dt$. Integrate the total-mass bound on $(0,1)$ and the inverse-power bound on $(1,\infty)$. □

Corollary 4 (sufficient sublinear spectral envelope). If there exist $0 < \alpha < 1$ and $C_\alpha < \infty$, independent of even N, such that

$$|e_j| \le C_\alpha(1 + |\lambda_j|^\alpha) \quad \text{for every reference eigenpair,} \tag{11}$$

then $|\operatorname{Tr}(S_N E_N)| = O_{\mu,\alpha}(1)$, and

$$\mathcal{D}_L(\mu) = \frac{\mu^2}{6}\log L + O_\mu(\sqrt{\log L}). \tag{12}$$

The envelope (11) is only a **sufficient, unresolved hypothesis**, and is weaker at large $|\lambda|$ than a logarithmic eigenpair envelope.

Alternative sufficient condition (uniform sublinearity). One can weaken (11) further: suppose that for each fixed $R < \infty$, the modular errors are uniformly bounded on $|\lambda_j| \le R$, and

$$\lim_{R\to\infty} \sup_{N \text{ even}, j:\,|\lambda_j|>R} \frac{|e_j|}{|\lambda_j|} = 0. \tag{13}$$

Then $\operatorname{Tr}(S_N E_N) = o(\log N)$ and the leading logarithmic law follows. Indeed, for any $\varepsilon > 0$, (13) bounds $|e_j| \le \varepsilon|\lambda_j|$ outside a fixed window while the window contributes $O(1)$. By layer cake and (9), $\sum_j q_j|\lambda_j| \le M_\mu + a_\mu \log N + b_\mu$, up to harmless fixed-size adjustments. Dividing the resulting estimate by $\log N$, taking the limsup, then letting $\varepsilon \downarrow 0$, proves the claim. Condition (13) is **not currently established**.

6. Scope and sources

Theorem 1, Corollary 2, and Lemma 3 are analytic consequences of the stated finite sine-kernel model and the referenced, separately established Jacobi/majorant results. They use no numerically inferred uniform limit. The longitudinal modular estimate is supplied separately in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*. None of these statements proves the proposed full finite part or the $O_\mu((\log L)^{-1/103})$ remainder.

Classical references: V. Eisler and I. Peschel, *Analytical Results for the Entanglement Hamiltonian of a Free-Fermion Chain*, J. Phys. A **50** (2017), 284003, doi:10.1088/1751-8121/aa76b5; D. Slepian, *Prolate Spheroidal Wave Functions, Fourier Analysis, and Uncertainty, V: The Discrete Case*, Bell Syst. Tech. J. **57** (1978), 1371–1430; V. Peller, *Hankel Operators and Their Applications*, Springer (2003), doi:10.1007/978-0-387-21681-2. These works furnish established mathematical context; they do not externally verify the Wilson-specific results presented here.

Contributions: Kirtus Justus directed the research; ChatGPT (OpenAI) developed the present weighted-variance derivation, checks and drafting. Claude (Anthropic), Gemini (Google), Grok (SpaceXAI), and DeepSeek (High-Flyer) are acknowledged for prior documented involvement in the broader research program, without attributing this particular derivation to them. No external human peer review or formal proof-assistant certification is claimed.

Sharp Logarithmic Jacobi Boundary Moment for a Half-Filled Fermion Interval

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Abstract

For the half-filled sine-kernel covariance on an interval of even length and its commuting tridiagonal concentration operator, we establish the sharp coefficient of the logarithmic weighted second moment. The proof uses an exact full-line boundary-kernel factorization and elementary lattice sums. We also give an exact finite-size expression involving the trigamma function, yielding a second independent route for numerical reconstruction. This result characterizes the size of a reference spectral weight needed in Wilson–Dirac relative-entropy estimates. It does not determine the remaining reference-modular correction and does not establish the complete relative-entropy logarithmic law.

1. Definitions and result

Let P be the orthogonal projection on $\ell^2(\mathbb{Z})$ with real Fourier kernel

$$T_d = \begin{cases} 1/2, & d = 0, \\ \sin(\pi d/2)/(\pi d), & d \neq 0. \end{cases}$$

Let χ_N be the projection onto sites $0, \dots, N-1$, and define $B_N = \chi_N P \chi_N$ on that interval. Define the Jacobi matrix J_N by zero diagonal and off-diagonal bonds $\alpha_j = j(N-j)/N$, $1 \le j < N$, with $\alpha_0 = \alpha_N = 0$. Put $W_N = B_N(I - B_N)$.

Theorem 1 (sharp second moment). As even $N \to \infty$,

$$\boxed{\operatorname{Tr}(J_N^2 W_N) = \frac{\log N}{3\pi^2} + O(1).}$$

The constant in the $O(1)$ term is absolute. Consequently, the $O(\log N)$ scale of this moment cannot be replaced by $o(\log N)$.

2. Exact Fourier boundary formula

Since $P^2 = P$,

$$W_N = \chi_N P(1-\chi_N)P\chi_N, \qquad \operatorname{Tr}(J_N^2 W_N) = \|(1-\chi_N)P\chi_N J_N\|_{\mathrm{HS}}^2.$$

For $0 \le j < N$ and $y \notin [0, N-1]$, the matrix element is

$$M_{yj} = \alpha_j T_{y-j+1} + \alpha_{j+1} T_{y-j-1}.$$

For $|y-j| = 1$, the boundary coefficients make this element zero. For odd separations $|y-j| \ge 3$, both kernel indices are nonzero even integers, so the element again vanishes. For even $d = y-j$,

$$|M_{yj}|^2 = \frac{[d(\alpha_j - \alpha_{j+1}) - (\alpha_j + \alpha_{j+1})]^2}{\pi^2(d^2-1)^2}.$$

Reflection about the midpoint of the interval preserves J_N and the even Fourier kernel T_d; hence the left and right exterior contributions have identical Hilbert–Schmidt norms.

For the left exterior write $y = -\ell - 1$, $\ell \geq 0$, and $n = j + \ell + 1$. Only even $n \geq 2$ contribute, and a direct substitution gives the *exact* numerator

$$d(\alpha_j - \alpha_{j+1}) - (\alpha_j + \alpha_{j+1}) = (\ell - j) - \frac{(2j+1)\ell + j}{N}. \tag{1}$$

Thus

$$\mathrm{Tr}(J_N^2 W_N) = \frac{2}{\pi^2} \sum_{\substack{n \geq 2 \\ n \text{ even}}} \frac{1}{(n^2-1)^2} \sum_{j=0}^{\min(n-1, N-1)} \left[(n-1-2j) - \frac{(2j+1)(n-j-1)+j}{N} \right]^2. \tag{2}$$

All terms are nonnegative, and this expression converges absolutely at each fixed N.

3. Extracting the logarithm with uniform remainder

For $2 \leq n \leq N$, let $u_{n,j} = n - 1 - 2j$ and $r_{n,j} = [(2j+1)(n-j-1)+j]/N$. Elementary arithmetic gives

$$\sum_{j=0}^{n-1} u_{n,j}^2 = \frac{n(n^2-1)}{3}, \qquad |u_{n,j}| \leq n, \qquad |r_{n,j}| \leq \frac{3n^2}{N}.$$

Using $(n^2-1)^2 \geq 9n^4/16$ for $n \geq 2$, the absolute difference between the terms from $(u-r)^2$ and from u^2, after division by $(n^2-1)^2$, is bounded by $C(1/N + n/N^2)$ for a universal constant C. Its sum over even $2 \leq n \leq N$ is $O(1)$.

For $n > N$, at most N values of j contribute. Here $0 \leq j < N$ and $\ell = n - 1 - j$, and the numerator in (1) is bounded in absolute value by $5n$. Hence the tail contributes at most

$$CN \sum_{n > N} n^{-2} = O(1).$$

The two exterior sides together therefore give

$$\mathrm{Tr}(J_N^2 W_N) = \frac{2}{3\pi^2} \sum_{\substack{2 \leq n \leq N \\ n \text{ even}}} \frac{n}{n^2-1} + O(1).$$

Writing $N = 2M$ and telescoping the partial fractions yields

$$\sum_{\substack{2 \leq n \leq N \\ n \text{ even}}} \frac{n}{n^2-1} = \frac{1}{2} \sum_{k=1}^{M} \left(\frac{1}{2k-1} + \frac{1}{2k+1} \right) = \frac{1}{2}[H_N + H_{N+1} - H_{N/2} - 1] = \frac{1}{2} \log N + O(1).$$

This proves Theorem 1. □

4. Exact finite-size evaluation with trigamma functions

Equation (2) also has a one-dimensional special-function form. For $0 \le j < N$, define

$$k_j = \left\lceil \frac{j+1}{2} \right\rceil, \quad a_j = 1 - \frac{2j+1}{N}, \quad b_j = -(2j+1) + \frac{2j^2+2j+1}{N}.$$

(The symbol b_j in this section denotes a polynomial coefficient, **not** a reference occupation number.) Put

$$C_j^- = \frac{1}{4}\psi_1(k_j - \tfrac{1}{2}), \qquad C_j^+ = \frac{1}{4}\psi_1(k_j + \tfrac{1}{2}), \qquad D_j = \frac{1}{2k_j - 1},$$

$$U_j = \frac{C_j^- + C_j^+ - D_j}{4}, \qquad V_j = \frac{C_j^- - C_j^+}{4}, \qquad R_j = \frac{D_j}{2} + U_j.$$

Then the following identity is exact for every positive even N:

$$\boxed{\mathrm{Tr}(J_N^2 W_N) = \frac{2}{\pi^2} \sum_{j=0}^{N-1} (a_j^2 R_j + 2a_j b_j V_j + b_j^2 U_j).} \tag{3}$$

Indeed, at fixed j, the numerator in (2) equals $a_j n + b_j$, with even $n \ge j+1$. The three series $\sum n^2/(n^2-1)^2$, $\sum n/(n^2-1)^2$, and $\sum 1/(n^2-1)^2$ evaluate respectively to R_j, V_j, U_j by partial fractions and the trigamma series. Formula (3) gives an independent finite-size reconstruction, without finite-matrix products or a Fourier quadrature cutoff.

5. Consequence and limitation for the Wilson relative-entropy problem

The known Wilson nonlinear correction may be bounded by pairing its transverse covariance component with a reference spectral weight containing $\mathrm{Tr}(J_N^2 W_N)$. Theorem 1 shows this moment is intrinsically logarithmic, not bounded. Thus the older second-moment-only Cauchy–Schwarz estimate gives $O_\mu(\sqrt{\log N})$, but the second moment alone cannot justify an improved scale. The separate *Fourth-Moment Control of the Wilson Nonlinear Correction*, sections 2–4, combines this moment with a positive fourth-moment bound to obtain a vanishing transverse contribution $O_\mu(\sqrt{(\log N)/N})$ and a bounded full nonlinear term. The nonlinear finite constant is obtained by Theorem 7 of *Finite Part of Wilson–Dirac Relative Entropy*. Theorem 6 of *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* improves the error to $O_\mu(\sqrt{(\log N)/N})$. This uses all-degree path control in addition to the moment bound proved here.

The separate term $\mathcal{F}_L = \mathrm{Tr}[(A_N - B_N)E_N]$ is bounded by the weighted theorem in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*. Together with the bounded nonlinear and entropy terms, that result establishes the complete leading law at fixed positive scaled mass. Neither the boundary moment alone nor equation (3) supplies this modular estimate. The complete fixed-positive-scaled-mass finite part and its $O_\mu(1/\log L)$ remainder are Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy*, using the separate modular, nonlinear and boundary estimates. The boundary moment alone does not imply them.

Literature

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Contributors and contributions

Kirtus Justus directed the research and is the sole human author. ChatGPT (OpenAI) assisted with derivations, computations and drafting. Claude (Anthropic), Gemini (Google), Grok (SpaceXAI), and DeepSeek (High-Flyer) are acknowledged for broader program participation, without unsupported attribution to this proof. No external peer review or institutional endorsement is claimed.

Half-Filled Sine-Kernel Endpoint and Extreme Modular Discrepancy

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Abstract

For the half-filled finite sine kernel and its commuting Jacobi matrix, we derive an exact integral formula and endpoint expansion for the reference scaling function, a Gaussian variational estimate for the largest Jacobi eigenvalue, and an exact Fourier-polynomial differential equation. The established fixed-index discrete-prolate concentration asymptotic determines the leading reference-modular discrepancy exactly, with limit $(\log \pi - 1)/2$. This is a fixed-mode conclusion and does not supply a uniform reference-modular estimate for the complete Wilson relative entropy.

1. Reference operators and conventions

Let $N = 2L$ and let the Hermitian $N \times N$ matrices B_N, J_N have entries

$$(B_N)_{jk} = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{i(j-k)\theta}\, d\theta, \qquad (J_N)_{j,j+1} = (J_N)_{j+1,j} = \frac{(j+1)(N-j-1)}{N},$$

for $0 \le j, k < N$, with all other entries of J_N zero. They commute, J_N has simple spectrum, and $0 < B_N < I$. Set

$$K_N = \log(I - B_N) - \log B_N, \qquad E_N = K_N + N\eta(J_N/N),$$

where

$$\eta(t) = \sum_{n=0}^{\infty} \frac{\pi \binom{2n}{n}\binom{4n}{2n}}{(2n+1)16^n} t^{2n+1}, \qquad |t| \le \tfrac{1}{2}.$$

The finite sine-kernel model specifies the reference operator and its commuting Jacobi operator.

2. Exact integral and endpoint expansion

Proposition 1. For $|t| \le 1/2$,

$$\boxed{\eta(t) = 4t \int_0^1 \frac{\sqrt{1-u^2}}{\sqrt{1-4t^2u^4}}\, du.} \qquad (2.1)$$

In particular,

$$\eta(1/2) = 2\,\mathrm{arsinh}\,1, \quad A_\eta := \max_{0<|t|\le 1/2} \frac{|\eta(t)|}{|t|} = 4\,\mathrm{arsinh}\,1,$$

and

$$C_\eta := \max_{0<|t|\le 1/2} \frac{|\eta(t)-\pi t|}{|t|^3} = 4(A_\eta - \pi).$$

Proof. Expand $(1-4t^2u^4)^{-1/2} = \sum_{n\ge 0}\binom{2n}{n}t^{2n}u^{4n}$. The beta integral gives

$$\int_0^1 u^{4n}\sqrt{1-u^2}\,du = \frac{\pi\binom{4n}{2n}}{4\,16^n(2n+1)},$$

which gives the stated coefficients. At the endpoint the majorant is integrable; monotone convergence applies to positive coefficients (the negative endpoint follows by oddness). For $t = 1/2$, cancellation of $1-u^2$ against $1-u^4$ gives $2\int_0^1 (1+u^2)^{-1/2}\,du$. The two maxima occur at the endpoint because the respective power series have nonnegative coefficients. □

Theorem 2. As $\delta \downarrow 0$,

$$\boxed{\eta'(1/2-\delta) = \frac{1}{\sqrt{2}}\log\frac{1}{\delta} + 2\sqrt{2}\log 2 + o(1),} \qquad (2.2)$$

$$\boxed{\eta(1/2-\delta) = 2\,\mathrm{arsinh}\,1 - \frac{\delta}{\sqrt{2}}\log\frac{1}{\delta} - \frac{1+4\log 2}{\sqrt{2}}\,\delta + o(\delta).} \qquad (2.3)$$

Proof. For $t < 1/2$ differentiation under the integral gives

$$\eta'(t) = 4\int_0^1 \frac{\sqrt{1-u^2}}{(1-4t^2u^4)^{3/2}}\,du.$$

Set $\varepsilon = 1-4t^2$, $y = 1-u^2$. Then

$$\eta'(t) = 2\int_0^1 \frac{\sqrt{y}\,dy}{\sqrt{1-y}\,[\varepsilon + (1-\varepsilon)(2y-y^2)]^{3/2}}.$$

Subtract the singular model

$$I_\varepsilon = 2\int_0^1 \frac{\sqrt{y}}{(\varepsilon+2y)^{3/2}}\,dy = \sqrt{2}\left[\mathrm{arsinh}\,\sqrt{2/\varepsilon} - \sqrt{\frac{2}{2+\varepsilon}}\right].$$

Its asymptotic is

$$I_\varepsilon = \frac{1}{\sqrt{2}}\log\frac{1}{\varepsilon} + \frac{3}{\sqrt{2}}\log 2 - \sqrt{2} + o(1).$$

Subtracting the two integrands before sending $\varepsilon \to 0$ leaves an integrable function, uniformly bounded at $y = 0$: expand the smooth factors near $y = 0$ and use $y^2 + \varepsilon y \lesssim (\varepsilon + y)^2$. The endpoint $y = 1$ is integrable. Dominated convergence therefore gives the limiting difference

$$R = 2\int_0^1 \frac{1}{y}\left[\frac{1}{\sqrt{1-y}(2-y)^{3/2}} - \frac{1}{2^{3/2}}\right]dy.$$

With $u = \sqrt{1-y}$, this becomes

$$R = \int_0^1 \left[\frac{4}{(1-u^2)(1+u^2)^{3/2}} - \frac{\sqrt{2}u}{1-u^2} \right] du.$$

The antiderivative, zero at $u = 0$, is

$$2\frac{u}{\sqrt{1+u^2}} + \sqrt{2}\,\operatorname{artanh}\frac{\sqrt{2}u}{\sqrt{1+u^2}} + \frac{1}{\sqrt{2}}\log(1-u^2).$$

The logarithms cancel at $u = 1$, giving $R = \sqrt{2} + (3/\sqrt{2})\log 2$. Consequently

$$\eta'(t) = \frac{1}{\sqrt{2}}\log\frac{1}{\varepsilon} + 3\sqrt{2}\log 2 + o(1).$$

Since $t = 1/2 - \delta$ entails $\varepsilon = 4\delta(1-\delta)$, this is (2.2). Integrating the derivative between t and $1/2$, and using the endpoint value, yields (2.3). □

3. Gaussian comparison for the top Jacobi eigenvalue

Theorem 3. Let λ_N be the maximum eigenvalue of J_N for even N. Then

$$\boxed{\lambda_N = \frac{N}{2} - \frac{1}{\sqrt{2}} + O(N^{-1}).} \qquad (3.1)$$

In addition, writing $\beta = \sqrt{2}$, every even $N \geq 2$ obeys the explicit bound

$$\boxed{\lambda_N \leq e^{-\beta/N}\left(\frac{N}{2} + \frac{6\cosh^2\beta}{5N}\right).} \qquad (3.2)$$

Proof. Write $x_j = j - (N-1)/2$ and choose $w_j = e^{-\beta x_j^2/N} > 0$. Collatz–Wielandt gives $\lambda_N \leq \max_j (J_N w)_j / w_j$. An exact calculation with $y = 2\beta x_j/N$ yields

$$\frac{(J_N w)_j}{w_j} = e^{-\beta/N}\left[\frac{N}{2}\left(1 - \frac{y^2}{2}\right)\cosh y - \frac{\cosh y}{2N} + \frac{y}{\beta}\sinh y\right].$$

The power series of $(1 - y^2/2)\cosh y$ has no quadratic term, a quartic coefficient $-5/24$, and strictly negative higher even coefficients, so the expression is at most $1 - 5y^4/24$. Also $|y| \leq \beta$ and $y\sinh y \leq y^2\cosh\beta$. Hence

$$\frac{(J_N w)_j}{w_j} \leq e^{-\beta/N}\left[\frac{N}{2} - \frac{5N}{48}y^4 + \frac{\cosh\beta}{\beta}y^2\right] \leq e^{-\beta/N}\left[\frac{N}{2} + \frac{6\cosh^2\beta}{5N}\right],$$

where the final step maximizes the displayed quadratic polynomial in $y^2 \geq 0$.

For the lower bound use the Rayleigh quotient of w. With $z_j = j + 1 - N/2$ for the bonds, an exact expression is

$$R_N = 2e^{-\beta/(2N)}\frac{\sum_{j=0}^{N-2}(N/4 - z_j^2/N)e^{-2\beta z_j^2/N}}{\sum_{j=0}^{N-1} e^{-2\beta x_j^2/N}} \leq \lambda_N.$$

Both sums are truncated Gaussian lattice sums. Extending to the full integer and half-integer lattices incurs exponentially small errors (up to polynomial prefactors), since the first omitted points have magnitude proportional to N. Poisson summation then gives

$$\sum_{k\in\mathbb{Z}} e^{-2\beta(k+\rho)^2/N} = \sqrt{\frac{\pi N}{2\beta}}\left[1 + O\left(e^{-\pi^2 N/(2\beta)}\right)\right], \qquad \rho \in \{0, 1/2\},$$

and the corresponding second Gaussian moment is $N/(4\beta)$ times the zeroth moment, up to exponentially small absolute relative errors. Therefore, for some $c > 0$,

$$R_N = e^{-\beta/(2N)}\left(\frac{N}{2} - \frac{1}{2\beta}\right) + O(e^{-cN}) = \frac{N}{2} - \frac{1}{\sqrt{2}} + \frac{3}{8N} + O(N^{-2}).$$

Together with (3.2), which equals $N/2 - 1/\sqrt{2} + O(N^{-1})$, this proves (3.1). The $3/(8N)$ coefficient in the *lower bound* is not being asserted as the next coefficient of the true eigenvalue. □

Corollary 4. Let $\lambda_N = \lambda_{\max}(J_N)$. Then

$$\boxed{N\eta(\lambda_N/N) = 2N \operatorname{arsinh} 1 - \frac{1}{2}\log N - \frac{1 + \log(16\sqrt{2})}{2} + o(1).} \qquad (3.3)$$

Proof. Set $\delta_N = 1/2 - \lambda_N/N = 1/(\sqrt{2}N) + O(N^{-2})$ in Theorem 2. The $O(N^{-2})$ input error gives $o(1)$ after multiplication by N, since η' grows only logarithmically. Simplification gives (3.3). □

4. An exact generating-polynomial and Fourier-leakage formulation

Theorem 5. Let $J_N v = \lambda v$ for a nonzero real vector $v = (v_0, \dots, v_{N-1})$ and define the polynomial $F(z) = \sum_{j=0}^{N-1} v_j z^j$. It satisfies the exact differential equation

$$\boxed{-z(1+z^2)F''(z) + [(N-1) + (N-3)z^2]F'(z) + [(N-1)z - N\lambda]F(z) = 0.} \qquad (4.1)$$

If $b = v^* B_N v/\|v\|^2$, then

$$\boxed{\frac{1-b}{b} = \frac{\int_{\pi/2<|\theta|\le\pi} |F(e^{i\theta})|^2\, d\theta}{\int_{-\pi/2}^{\pi/2} |F(e^{i\theta})|^2\, d\theta}.} \qquad (4.2)$$

For the top eigenpair, v may be chosen strictly positive and symmetric under $j \mapsto N-1-j$; consequently $F(z) = z^{N-1}F(1/z)$.

Proof. The recurrence $N\lambda v_j = (j+1)(N-j-1)v_{j+1} + j(N-j)v_{j-1}$ is understood with $v_{-1} = v_N = 0$. Multiply by z^j and sum over j. The shifted sums give $(N-1)F' - zF''$ and $(N-1)zF + (N-3)z^2F' - z^3F''$, respectively, proving (4.1). Equation (4.2) follows directly from the Fourier projection formula for B_N and Parseval. Positivity and strict Perron simplicity hold because all Jacobi off-diagonal entries are positive; reflection commutes with J_N, hence the top vector is symmetric. □

Fourier-leakage identity and endpoint criterion. Let v_N be the top normalized Jacobi eigenvector, and write I_N^{in} and I_N^{out} for the two integrals in (4.2). Its scalar modular discrepancy is

$$e_N^{\text{edge}} = \log\frac{I_N^{\text{out}}}{I_N^{\text{in}}} + N\eta(\lambda_N/N).$$

By (3.3), **boundedness of this one edge discrepancy** is equivalent to

$$\boxed{\frac{I_N^{\text{out}}}{I_N^{\text{in}}} = \sqrt{N}\, e^{-2N \operatorname{arsinh} 1 + O(1)}.} \qquad (4.3)$$

The $O(1)$ is in the logarithm and means two-sided multiplicative bounds independent of N. The fixed-index concentration asymptotic in the next section establishes this criterion for the leading mode. It does not provide uniform control over eigenvalue indices that grow with N.

5. Extreme modular discrepancy from classical discrete-prolate concentration

Theorem 6 (application of the classical discrete-prolate fixed-index asymptotic). In the preceding half-filled normalization,

$$\boxed{1 - b_N \sim \sqrt{\pi}\, 2^{9/4} \sqrt{N}\, e^{-2N \operatorname{arsinh}(1)}.} \qquad (5.1)$$

Consequently,

$$\boxed{\lim_{\substack{N\to\infty \\ N \text{ even}}} e_N = \frac{\log \pi - 1}{2}.} \qquad (5.2)$$

In particular, the discrepancy for the **single top Jacobi eigenmode** is bounded uniformly over even $N \geq 2$.

Proof. The classical discrete-prolate concentration matrix is

$$P(N, W)_{jk} = \begin{cases} \frac{\sin(2\pi W(j-k))}{\pi(j-k)}, & j \neq k, \\ 2W, & j = k. \end{cases}$$

It equals B_N for $W = 1/4$. The fixed-index eigenvalue asymptotic of Slepian (1978), explicitly transcribed in Adcock, Huybrechs and Martín-Vaquero, *On the numerical stability of Fourier extensions*, §3.2, equation (3.10), states for $k = 0$,

$$1 - b_N \sim \sqrt{\pi}\, 2^{9/4} \alpha^{1/4} (2-\alpha)^{-1/2} \sqrt{N}\, \beta^{-N}, \quad \alpha = 1 - \cos(2\pi W), \quad \beta = \frac{\sqrt{2} + \sqrt{\alpha}}{\sqrt{2} - \sqrt{\alpha}}.$$

At $W = 1/4$, $\alpha = 1$, while $\beta = 3 + 2\sqrt{2} = \exp(2 \operatorname{arsinh} 1)$. This proves (5.1). Since $b_N \to 1$, its odds satisfy

$$\log \frac{1 - b_N}{b_N} = -2N \operatorname{arsinh}(1) + \frac{1}{2} \log N + \log(\sqrt{\pi} 2^{9/4}) + o(1).$$

Adding the Jacobi-edge expansion (3.3) and simplifying the constants proves (5.2). A convergent sequence is bounded for sufficiently large N; the finitely many remaining even N have finite discrepancies since $0 < b_N < 1$. □

Source attribution. The concentration-eigenvalue asymptotic used here is due to Slepian and is reproduced by Adcock, Huybrechs and Martín-Vaquero. This is an application of their theorem, not an independent proof of it. The source's theorem applies to fixed eigenvalue index and fixed bandwidth; no uniformity in an index growing with N has been imported. The proof of Slepian's original asymptotic is not reproduced here.

6. Exact real angular equation

Let $J_N v = \lambda v$, $F(z) = \sum_{j=0}^{N-1} v_j z^j$, and $\phi(\theta) = e^{-i(N-1)\theta/2} F(e^{i\theta})$. Summing the Jacobi recurrence gives

$$-z(1+z^2)F'' + [(N-1) + (N-3)z^2]F' + [(N-1)z - N\lambda]F = 0.$$

Proposition 7. Every generating polynomial satisfies

$$\boxed{\frac{d}{d\theta}(\cos\theta\,\phi'(\theta)) + \left[\frac{N^2-1}{4}\cos\theta - \frac{N\lambda}{2}\right]\phi(\theta) = 0.} \qquad (6.1)$$

Proof. On the unit circle $z\partial_z = -i\partial_\theta$. Substitute $F(e^{i\theta}) = e^{i(N-1)\theta/2}\phi(\theta)$ into the displayed polynomial equation, collect terms in ϕ'', ϕ', and ϕ, and divide by two. The resulting expression is (6.1). This algebra is independent of any asymptotic approximation. □

For the top eigenvector, reflection symmetry makes ϕ real and even. For even N, palindromicity of odd polynomial degree implies $\phi(\pi) = 0$, and (6.1) at the band edge gives

$$\phi'(\pi/2) = -\frac{N\lambda_N}{2}\phi(\pi/2). \qquad (6.2)$$

Multiplying (6.1) by ϕ and integrating by parts gives the exact exterior identity

$$\boxed{\int_{\pi/2}^{\pi} |\cos\theta|\,\phi'(\theta)^2\,d\theta = \int_{\pi/2}^{\pi}\left[\frac{N^2-1}{4}|\cos\theta| + \frac{N\lambda_N}{2}\right]\phi(\theta)^2\,d\theta.} \qquad (6.3)$$

The analogous interior formula over $[0, \pi/2]$ has $\cos\theta$ instead of its absolute value and the minus sign preceding $N\lambda_N/2$. Neither identity supplies a two-sided exterior mass estimate by itself.

7. Exact action and asymptotic interpretation

The leading interior normal-form action computed from (6.1) is

$$\mathcal{A} = \frac{1}{2}\int_0^{\pi/2} \sqrt{\sec\theta - 1}\,d\theta = \operatorname{arsinh}(1).$$

Indeed, with $t = \tan(\theta/2)$ and then $u = \sqrt{1-t^2}$, the integral becomes $\sqrt{2}\int_0^1 (2-u^2)^{-1}du = \operatorname{artanh}(1/\sqrt{2}) = \operatorname{arsinh}(1)$.

This matches the exponential action in the classical theorem (5.1). It is an exact evaluation of a leading formal action, **not** a standalone rigorous derivation of the prolate eigenvalue prefactor. In particular, the behavior at the singular band edge and the central turning point requires controlled connection formulas if one seeks a direct ODE proof.

8. Scope and limitations

The fixed-index theorem controls only the highest Jacobi eigenvector and its paired sine-kernel occupation. It cannot be extended to an index growing with N without new estimates. The commuting sine-kernel/Jacobi spectrum includes central, transitional, and other edge regions that are not bounded by the single-mode result.

For the Wilson–Dirac interval problem the established reduction is

$$\mathcal{D}_L(\mu) = \frac{\mu^2}{6}\log L + \operatorname{Tr}(S_N E_N) + O_\mu(\sqrt{\log L}), \qquad N = 2L.$$

Neither the scalar endpoint expansion nor the Slepian fixed-index result alone establishes the leading coefficient or finite part of the complete relative entropy. The separate argument in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* gives $\operatorname{Tr}(S_N E_N) = O_\mu(1)$ and the leading entropy coefficient; it does not establish the entropy finite part or an expansion uniform up to the spectral edge.

References

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Kirtus Justus directed the research and determined the scientific scope. ChatGPT (OpenAI) derived and checked the scalar endpoint and Jacobi spectral-edge results, identified the precise classical discrete-prolate theorem for the fixed-index concentration asymptotic, and prepared the mathematical synthesis. The Slepian asymptotic is attributed to its original source; it is not claimed as an original result. Claude (Anthropic), Gemini (Google), Grok (SpaceXAI), and DeepSeek (High-Flyer) are acknowledged for broader research participation without attributing verification of these calculations to them. No external peer review, degree, academic affiliation or institutional endorsement is claimed.

Fixed-Index Jacobi Spectral Expansion Through Inverse-Square Order

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Abstract

For the tridiagonal Jacobi matrix commuting with the half-filled finite sine-kernel concentration operator, we prove an expansion of each fixed-index upper-edge eigenvalue through the inverse-square term, with an index-dependent inverse-cube error. The argument begins with an exact finite-dimensional quadratic form and combines compact spectral convergence to a harmonic oscillator, quantitative weighted Taylor bounds for sampled Hermite functions, and two perturbative corrections to a finite Hermite quasimode. The spectral-gap argument also controls the approximation of the corresponding fixed-index eigenvector. The scope is fixed index. The theorem does not control indices growing with the matrix size, reference-modular operator norms, or Wilson–Dirac relative-entropy asymptotics.

1. Exact problem and result

For an even positive integer N, let J_N be the real symmetric $N \times N$ matrix

$$(J_N)_{j,j+1} = (J_N)_{j+1,j} = \frac{(j+1)(N-j-1)}{N}, \qquad 0 \le j \le N-2,$$

with all diagonal entries zero. The off-diagonal entries are strictly positive, so the eigenvalues are simple. Write them in strictly decreasing order as $\lambda_{N,0} > \cdots > \lambda_{N,N-1}$.

Theorem. For each fixed nonnegative integer k, as $N \to \infty$ through even integers,

$$\boxed{\lambda_{N,k} = \frac{N}{2} - \frac{2k+1}{\sqrt{2}} + \frac{6k^2+6k+3}{8N} + \frac{\sqrt{2}(2k+1)(k^2+k+5)}{64N^2} + O_k(N^{-3}).} \tag{1}$$

The remainder constant and the starting dimension may depend on k. All statements below use a fixed k; no uniformity over moving indices is asserted.

2. Discrete form and identification of the oscillator index

Set $h = N^{-1/2}$, $H_N = (N/2)I - J_N$, $y_j = h(j - (N-1)/2)$, and

$$a_{j+1/2} = \frac{1}{4h^2} - (y_j + h/2)^2 = \frac{(j+1)(N-j-1)}{N}.$$

The exterior coefficients $a_{-1/2}$ and $a_{N-1/2}$ vanish exactly. Summing squares gives the exact formula

$$\boxed{\langle v, H_N v\rangle = \sum_{j=0}^{N-2} a_{j+1/2}|v_{j+1} - v_j|^2 + \sum_{j=0}^{N-1}(2y_j^2 + h^2/2)|v_j|^2.} \tag{2}$$

The limiting oscillator is

$$H = -\tfrac{1}{4}\partial_y^2 + 2y^2, \qquad H\psi_k = \varepsilon_k \psi_k, \qquad \varepsilon_k = (2k+1)/\sqrt{2}, \tag{3}$$

with normalized Hermite functions ψ_k. We justify the spectral numbering instead of deriving it from a quasimode alone.

For $u \in \mathcal{S}(\mathbb{R})$, define $(\mathcal{S}_h u)_j = \sqrt{h}\, u(y_j)$. Riemann-sum convergence gives convergence of scalar products and of (2) on any fixed finite span of Hermite functions. The min–max principle therefore yields

$$\limsup_{N\to\infty} \varepsilon_{N,k} \le \varepsilon_k, \qquad \varepsilon_{N,k} = N/2 - \lambda_{N,k},$$

where the discrete energies are ordered increasingly.

For the converse, take normalized vectors $v^{(N)}$ with uniformly bounded energy. Formula (2) implies $\sum_j y_j^2 |v_j^{(N)}|^2 \le C$, so their mass outside $|y| \le R$ is at most C/R^2, uniformly in N. Embed them into $L^2(\mathbb{R})$ by the piecewise constant functions $v_j/\sqrt{h}$ on the cells centered at y_j. These embeddings preserve the ℓ^2 norm exactly. For the bond midpoint $y_{j+1/2} = y_j + h/2$, the coefficient satisfies the quantitative local bound

$$a_{j+1/2} = \frac{1}{4h^2} - y_{j+1/2}^2 \ge \frac{1}{8h^2}, \qquad |y_{j+1/2}| \le R, \quad h^2R^2 \le \tfrac{1}{8}.$$

Consequently, for discrete vectors of energy at most C, $\sum_{|y_{j+1/2}|\le R} |v_{j+1} - v_j|^2/h^2 \le 8C$. The piecewise linear interpolants therefore have uniformly bounded local H^1 seminorms, and their local L^2 distance from the piecewise constant embeddings is $O_R(h)$. Local Rellich compactness plus the uniform tail control gives global strong L^2 precompactness. By lower semicontinuity and the local limit of the bond coefficients, every limit u satisfies

$$\langle u, Hu\rangle \le \liminf_N \langle v^{(N)}, H_N v^{(N)}\rangle$$

in the quadratic-form sense. Apply this to the first $k+1$ discrete energy eigenvectors simultaneously, including all their fixed linear combinations. Their limits remain orthonormal. Continuum min–max then gives $\liminf_N \varepsilon_{N,k} \ge \varepsilon_k$. Thus

$$\boxed{\varepsilon_{N,k} \to \varepsilon_k \quad \text{for every fixed } k.} \tag{4}$$

In particular, each sufficiently small fixed neighborhood of ε_k contains precisely one discrete eigenvalue for sufficiently large N. This fact will identify the eigenvalue located by the quasimode.

3. Uniform Taylor remainder on fixed Hermite spans

For a smooth function sampled at $y = y_j$, the exact finite action is

$$\begin{aligned}(H_N \mathcal{S}_h u)_j/\sqrt{h} = {} & (2y^2 + h^2/2)u(y) \\ & - \left(\frac{1}{4h^2} - y^2 - \frac{h^2}{4}\right)\big[u(y+h) + u(y-h) - 2u(y)\big] \\ & + yh\big[u(y+h) - u(y-h)\big].\end{aligned} \tag{5}$$

This identity holds even at an endpoint, because the coefficient multiplying the exterior value is identically zero.

Define

$$
\begin{aligned}
P &= \partial_y(y^2\partial_y) + \tfrac{1}{2} - \tfrac{1}{48}\partial_y^4, \\
Q &= \tfrac{1}{12}\partial_y^2(y^2\partial_y^2) + \tfrac{1}{12}\partial_y^2 - \tfrac{1}{1440}\partial_y^6.
\end{aligned}
\tag{6}
$$

Taylor lemma. For each fixed finite Hermite span $\mathcal{H}_M = \mathrm{span}\{\psi_0, \dots, \psi_M\}$, and for each fixed $u \in \mathcal{H}_M$,

$$
\|H_N \mathcal{S}_h u - \mathcal{S}_h(Hu + h^2 Pu + h^4 Qu)\|_{\ell^2} \le C_u h^6. \tag{7}
$$

The estimate also holds uniformly on the unit sphere of any fixed finite Hermite span, after enlarging the constant.

Justification. Taylor's theorem with integral remainders gives

$$
\begin{aligned}
u(y+h) + u(y-h) - 2u(y) &= h^2 u'' + \tfrac{h^4}{12}u^{(4)} + \tfrac{h^6}{360}u^{(6)} + O_u(h^8), \\
u(y+h) - u(y-h) &= 2hu' + \tfrac{h^3}{3}u^{(3)} + O_u(h^5),
\end{aligned}
$$

where the remainders admit bounds by finite sums of Schwartz seminorms, including the y^2 weights in (5). The leading coefficient $1/(4h^2)$ turns the h^8 symmetric-difference remainder into $O(h^6)$; the remaining terms are also $O(h^6)$ or smaller in the sampled ℓ^2 norm. The weighted sampled Schwartz norms stay bounded uniformly in h by Riemann-sum estimates. Equating the h^0, h^2, h^4 terms gives exactly H, P, Q. The endpoint treatment in (5) requires no nonphysical boundary value: the combined coefficients of the exterior samples vanish exactly. The discarded Gaussian tails are smaller than any inverse power of N. The same argument may be taken one order further: on fixed Hermite spans,

$$
H_N \mathcal{S}_h u = \mathcal{S}_h\big[(H + h^2 P + h^4 Q + h^6 R_6)u\big] + O_u(h^8),
$$

where the additional *Taylor remainder operator*, used only to cross-check the residual scale, is

$$
R_6 = \frac{y^2}{360}\partial_y^6 + \frac{y}{60}\partial_y^5 + \frac{1}{48}\partial_y^4 - \frac{1}{80640}\partial_y^8.
$$

Every derivative in this display acts on a fixed Schwartz function and its sampled weighted norms are uniformly bounded. This higher expansion does not change the statement of the spectral theorem. □

4. Two perturbative coefficients

Use the Hermite ladder operators with $\alpha = 2\sqrt{2}$:

$$
y = \frac{a + a^\dagger}{\sqrt{2\alpha}}, \qquad \partial_y = \sqrt{\frac{\alpha}{2}}(a - a^\dagger).
$$

For any $k \ge 0$, negative-index terms being absent, the nonzero matrix elements of $P\psi_k$ are

$$
\begin{aligned}
P_{kk} &= -\frac{6k^2+6k+3}{8} =: p_k,\\
P_{k-2,k} &= \frac{(2k-1)\sqrt{k(k-1)}}{12},\\
P_{k+2,k} &= \frac{(2k+3)\sqrt{(k+1)(k+2)}}{12},\\
P_{k-4,k} &= \frac{5}{24}\sqrt{k(k-1)(k-2)(k-3)},\\
P_{k+4,k} &= \frac{5}{24}\sqrt{(k+1)(k+2)(k+3)(k+4)}.
\end{aligned}
\tag{8}
$$

The second diagonal coefficient is

$$
\langle \psi_k, Q\psi_k\rangle = \frac{\sqrt{2}}{36}(4k^3+6k^2+8k+3). \tag{9}
$$

Introduce the finite correction

$$
w_k = \sum_{\ell\neq k} \frac{P_{\ell k}}{\varepsilon_k-\varepsilon_\ell}\psi_\ell, \quad (H-\varepsilon_k)w_k = -(P-p_k)\psi_k.
$$

The inverse-square energy coefficient equals

$$
\begin{aligned}
q_k &= \langle \psi_k, Q\psi_k\rangle + \sum_{\ell\neq k}\frac{|P_{\ell k}|^2}{\varepsilon_k-\varepsilon_\ell}\\
&= -\frac{\sqrt{2}(2k+1)(k^2+k+5)}{64}.
\end{aligned}
\tag{10}
$$

Only the four possible off-diagonal shifts in (8) enter the sum; its simplification is an identity in k. No infinite perturbation sum is required.

5. Second corrected quasimode and proof of the theorem

The combination

$$
R_k = (P-p_k)w_k + (Q-q_k)\psi_k
$$

has zero component on ψ_k, by (10), and belongs to a finite Hermite span. Consequently the finite vector

$$
z_k = -\sum_{\ell\neq k}\frac{\langle \psi_\ell, R_k\rangle}{\varepsilon_\ell-\varepsilon_k}\psi_\ell
$$

satisfies $(H-\varepsilon_k)z_k = -R_k$. Let

$$
U_{N,k} = \mathcal{S}_h(\psi_k + h^2 w_k + h^4 z_k), \qquad \mathcal{E}_{N,k} = \varepsilon_k + h^2 p_k + h^4 q_k.
$$

Insert these expressions into (7). The coefficients of h^0, h^2, h^4 cancel separately, leaving

$$
\|(H_N - \mathcal{E}_{N,k})U_{N,k}\|_{\ell^2} \le C_k h^6, \qquad \|U_{N,k}\|_{\ell^2} \longrightarrow 1. \tag{11}
$$

For a self-adjoint finite-dimensional matrix, a normalized vector with residual at most ρ places an eigenvalue within ρ of its proposed energy. Apply this to (11), and use the indexed convergence (4) and oscillator spectral gaps to identify the nearby eigenvalue as $\varepsilon_{N,k}$. In fact, the same spectral gap yields control of the eigenvector itself: if $v_{N,k}$ is its normalized eigenvector and $\widehat{U}_{N,k} = U_{N,k}/\|U_{N,k}\|$, then there is a choice of sign such that

$$\min_{\sigma\in\{+1,-1\}} \|\widehat{U}_{N,k} - \sigma v_{N,k}\|_{\ell^2} = O_k(h^6).$$

To see this, use the convergent separation of the adjacent discrete energies to bound the spectral projection onto the orthogonal complement of $v_{N,k}$ by the residual in (11) divided by a fixed positive gap. This statement has fixed-index scope only. It follows that

$$\varepsilon_{N,k} = \varepsilon_k + h^2 p_k + h^4 q_k + O_k(h^6).$$

Since $h^2 = N^{-1}$ and $\lambda_{N,k} = N/2 - \varepsilon_{N,k}$, equation (1) follows. □

6. Relationship to the sine-kernel reference and its limits

The half-filled finite sine kernel has entries

$$(B_N)_{j\ell} = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} e^{i(j-\ell)\theta}\, d\theta.$$

For $W = 1/4$, the reference matrix B_N is Slepian's finite concentration matrix $p(N, W)$, equation (21), and $J_N = \frac{2}{N}\sigma(N, 1/4)$ in terms of his commuting tridiagonal matrix, equation (14). Slepian proves that the ordered eigenvectors of $\sigma(N, W)$ have strictly decreasing finite concentration eigenvalues, §4.1, equations (129)–(131), pp. 1403–1404. The argument uses Sturm zero ordering and continuity in the bandwidth; **commutation alone would not imply the index matching**. Thus the k-th largest Jacobi eigenvector corresponds to the k-th largest occupation of B_N. With this classical ordering input, Slepian's fixed-index concentration asymptotic (reproduced by Adcock, Huybrechs and Martín-Vaquero, §3.2, equation (3.10)) and the scalar endpoint expansion give, for fixed k, the modular residual limit

$$\lim_{N\to\infty} e_{N,k} = \frac{1}{2}\log\pi - \log(k!) - k\log 2 + \left(k + \tfrac{1}{2}\right)[\log(2k+1) - 1].$$

The extra inverse-square Jacobi coefficient in (1) **does not** imply a next-order modular residual: a compatible quantitative correction to the discrete-prolate concentration asymptotic would also be required. Nothing in this manuscript establishes a bound uniform for $k = k(N)$, the operator-norm bound $\|E_N\| = O(1)$, or the required Wilson longitudinal modular trace estimate $\mathrm{Tr}(S_N E_N) = o_\mu(\log N)$. The separate whole-line argument in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* establishes the fixed-positive-mass leading entropy law with bounded remainder. The fixed-index analysis here does not supply the whole-line argument. Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy* determines the finite part for fixed $\mu > 0$, and Theorem 3 of *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* proves $\|E_N\| \le 7$ for even $N \ge 2$. Neither statement turns this fixed-index expansion into a moving-index expansion.

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Finite-Interval Occupation Logarithm and Reference-Crossing Bounds

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Abstract

We establish a sharp scalar occupation-logarithm inequality, an explicit logarithmic envelope for the half-filled sine-kernel crossing nuclear norm, and finite-dimensional trace consequences for Wilson–Dirac interval covariances. An elementary change of variables and a two-region majorization yield an explicit crossing bound with leading coefficient one-half in its logarithmic integral estimate. Weighted Hilbert–Schmidt geometry gives a companion inequality for the reference modular discrepancy. These inequalities refine the constants in a directional positive-modular-trace enclosure and a transverse reference estimate. The Wilson positive reference-modular trace asymptotic requires the separate weighted argument of *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*.

1. Reference operators and hypotheses

Let $N = 2L \geq 2$ be even. Denote by B_N the half-filled sine-kernel compression, J_N its commuting Jacobi operator, and

$$W_N = B_N(I - B_N), \qquad K_N = \log(I - B_N) - \log B_N, \qquad E_N = K_N + N\eta(J_N/N).$$

The finite spectrum of B_N lies strictly in $(0, 1)$, and B_N and J_N commute. For the longitudinal and positive-majorant operators S_N, G_N, use the Wilson full-line relations

$$0 \leq 2G \leq I, \quad [G, P_0] = 0, \quad S = G(I - 2P_0)$$

and their interval compressions. The finite particle–hole staggering D satisfies $DB_ND = I - B_N$, $DG_ND = G_N$, $DS_ND = -S_N$. In this geometry the finite even-dimensional reference has no occupation eigenvalue $1/2$.

Write

$$M_N = \operatorname{Tr} G_N, \quad M_{2,N} = \operatorname{Tr}(W_N J_N^2), \quad T_{K,N} = \operatorname{Tr}(W_N K_N^2).$$

The Wilson-majorant and reference results used below are the finite mass identity

$$M_N = N\left[\frac{1}{2} - \frac{\arccos r}{\pi\sqrt{1 - r^2}}\right], \qquad r = \frac{\mu}{N + \mu}, \qquad M_N \leq \frac{\mu}{\pi},$$

and the reference estimates

$$\operatorname{Tr}\sqrt{W_N} \leq \frac{2}{\pi} I_N, \quad I_N = \int_0^\infty \frac{e^{-t}\sqrt{1 - e^{-2Nt}}}{1 - e^{-2t}}\, dt, \quad M_{2,N} \leq \frac{32}{\pi^2}(H_N + 1),$$

where $H_N = \sum_{j=1}^{N} j^{-1}$. These imported Wilson-specific inputs require their own separately established proofs; no step below substitutes an external classical reference for them.

2. Optimal occupation-logarithm constant

Theorem 1. There is a unique $x_* > 0$ with $x_* \tanh x_* = 2$. The best universal constant in

$$b(1-b)\log^2\frac{1-b}{b} \le C_{\log}\sqrt{b(1-b)}, \qquad 0 < b < 1,$$

is

$$\boxed{C_{\log} = \frac{2x_*^2}{\cosh x_*} = 2.12888831330963714924809207535747 6\ldots.}$$

Proof. With $x = \frac{1}{2}|\log((1-b)/b)|$, the ratio of the left side to $\sqrt{b(1-b)}$ is $2x^2/\cosh x$. Its derivative has sign $2-x\tanh x$. The latter expression changes sign exactly once because $x\tanh x$ is strictly increasing from zero to infinity. The ratio vanishes at both endpoints of the x-axis, and the unique maximum is attained by the two particle–hole conjugate occupations. □

Therefore, by the spectral theorem,

$$\boxed{T_{K,N} \le C_{\log}\operatorname{Tr}\sqrt{W_N}.} \tag{1}$$

3. Explicit reference-crossing integral

Theorem 2. For every integer $N \ge 2$,

$$\boxed{I_N \le F_N := \operatorname{arsinh}\sqrt{N-1} + \sqrt{N}\arcsin(N^{-1/2}).} \tag{2}$$

In particular,

$$F_N = \tfrac{1}{2}\log N + 1 + \log 2 + O(N^{-1}). \tag{3}$$

Proof. Set $u = e^{-2t}$ in the defining integral to obtain the exact identity

$$I_N = \frac{1}{2}\int_0^1 \frac{\sqrt{1-u^N}}{\sqrt{u}(1-u)}\,du.$$

Let $a = 1 - N^{-1}$. On $(0,a)$, $\sqrt{1-u^N} \le 1$. On $(a,1)$, Bernoulli's inequality $u^N \ge 1 - N(1-u)$ gives $\sqrt{1-u^N} \le \sqrt{N(1-u)}$. Consequently,

$$\begin{aligned} I_N &\le \frac{1}{2}\int_0^a \frac{du}{\sqrt{u}(1-u)} + \frac{\sqrt{N}}{2}\int_a^1 \frac{du}{\sqrt{u(1-u)}} \\ &= \operatorname{artanh}\sqrt{1-N^{-1}} + \sqrt{N}\arcsin(N^{-1/2}) \\ &= F_N. \end{aligned}$$

Expanding the two elementary functions at large N gives (3). □

For a completely explicit comparison with a logarithmic envelope, note that $\operatorname{arsinh}\sqrt{N-1} \le \frac{1}{2}\log N + \log 2$, and the function $\arcsin x/x$ is increasing for $0 < x < 1$. Consequently, for $N \ge 2$,

$$F_N \le \tfrac{1}{2}\log N + \log 2 + \frac{\pi}{2\sqrt{2}} < \frac{\log N + 2\sqrt{2} + e^{-1}}{1 - e^{-2}}.$$

The displayed inequalities are uniform in N and do not claim optimality of the integral majorant.

Corollary 3. Every finite interval satisfies

$$\operatorname{Tr}\sqrt{W_N} \le \frac{2F_N}{\pi}, \qquad T_{K,N} \le \frac{2C_{\log}F_N}{\pi}. \tag{4}$$

This does not assert asymptotic equality; the upper bound carries no inferred lower-bound coefficient.

4. Directional Wilson positive-trace enclosure

Let $P_N = \mathrm{Tr}(G_N|K_N|)$. The full-line projector relations give $0 \le G + S \le 2G$ and $G + S \le I - P_0$, and the inequalities persist under interval compression. For a reference eigenvector with occupation $b > 1/2$, denote $q = \langle v, G_N v\rangle$ and $s = \langle v, S_N v\rangle$. Then $u = q + s$ obeys $0 \le u \le 2q$ and $u \le 1 - b$. Particle–hole pairing yields

$$P_N - \mathrm{Tr}(S_N K_N) = 2 \sum_{b_j > 1/2} u_j |k_j| \ge 0.$$

Since $u_j \le \sqrt{2q_j(1-b_j)}$, $\sum_{b_j>1/2} q_j = M_N/2$, $b_j \ge 1/2$, and $T_{K,N} = 2\sum_{b_j>1/2} b_j(1-b_j)k_j^2$, Cauchy–Schwarz proves

$$\boxed{0 \le P_N - \mathrm{Tr}(S_N K_N) \le U_N := 2\sqrt{M_N T_{K,N}}.} \tag{5}$$

Corollary 4 (fully explicit error). With $C_{\log}$ from Theorem 1 and F_N from Theorem 2,

$$\boxed{U_N \le U_{N,\mu}^{\mathrm{elem}} := 2\sqrt{\frac{2C_{\log} M_N F_N}{\pi}}.} \tag{6}$$

For fixed positive μ, the bound has the asymptotic upper-envelope behavior

$$U_{N,\mu}^{\mathrm{elem}} = \frac{2}{\pi}\sqrt{C_{\log}\mu \log N}\,[1 + o(1)]. \tag{7}$$

The factor $O(\sqrt{\log N})$ is not improved to $O(1)$. Combining (5) with the independently proved cubic Jacobi envelope gives, with $C_\eta = 16\,\mathrm{arsinh}(1) - 4\pi$, $a_\mu = \sqrt{2/15}\,\mu^2/\pi$, $b_\mu = \mu/(3\pi)$,

$$V_{N,\mu} = C_\eta \left(\frac{3a_\mu}{8} + \frac{3b_\mu}{2N}\right),$$

$$\boxed{P_N + \pi\,\mathrm{Tr}(S_N J_N) - U_{N,\mu}^{\mathrm{elem}} - V_{N,\mu} \le \mathrm{Tr}(S_N E_N) \le P_N + \pi\,\mathrm{Tr}(S_N J_N) + V_{N,\mu}.} \tag{8}$$

5. Weighted reference-modular variance

Let $A_\eta = 4\,\mathrm{arsinh}(1)$, so the established scaling-function bound is $|\eta(t)| \le A_\eta |t|$ for $|t| \le 1/2$. Because B_N, J_N, and their functions commute, the weighted Hilbert–Schmidt triangle inequality gives

$$\begin{aligned}\sqrt{\mathrm{Tr}(W_N E_N^2)} &= \|\sqrt{W_N}(K_N + N\eta(J_N/N))\|_2 \\ &\le \sqrt{T_{K,N}} + \sqrt{\mathrm{Tr}\{W_N[N\eta(J_N/N)]^2\}} \\ &\le \sqrt{C_{\log}\,\mathrm{Tr}\,\sqrt{W_N}} + A_\eta\sqrt{M_{2,N}}.\end{aligned}$$

Corollary 5. Consequently,

$$\boxed{\sqrt{\mathrm{Tr}(W_N E_N^2)} \le \sqrt{\frac{2C_{\log}F_N}{\pi}} + \frac{4\sqrt{2}A_\eta}{\pi}\sqrt{H_N + 1}.} \tag{9}$$

The established transverse pairing estimate $|\langle v_j, D_N v_j\rangle| \le 2\sqrt{q_j b_j(1-b_j)}$ now gives

$$\boxed{|\mathrm{Tr}(D_N E_N)| \le 2\sqrt{M_N}\left[\sqrt{\frac{2C_{\log}F_N}{\pi}} + \frac{4\sqrt{2}A_\eta}{\pi}\sqrt{H_N+1}\right].} \tag{10}$$

Both results are still $O_\mu(\sqrt{\log N})$. They rely on the Wilson pairing, crossing and second-moment theorems, not on a pointwise finite-size approximation of individual modular eigenvalues.

6. Scientific scope

The leading-order Wilson reduction and the exact Jacobi first moment imply the reformulation

$$\mathcal{D}_L(\mu) = P_N + \frac{\mu^2}{4} - \frac{\mu^2}{6}\log\frac{8}{\mu} + O_\mu(\sqrt{\log N}), \qquad N = 2L.$$

The finite-dimensional bounds here do not determine the limit of $P_N/\log N$. The separate leading entropy law of *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*, combined with the positive-trace reduction above, establishes

$$\boxed{P_N = \frac{\mu^2}{6}\log N + o_\mu(\log N).}$$

The estimates established here concern occupation sums and reference crossing. The finite-part limit and uniform modular bound are established in *Finite Part of Wilson–Dirac Relative Entropy* and *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals*, respectively. Their results do not assert a moving-edge interpolation expansion.

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- Companion mathematical manuscripts for the Wilson-specific majorant, reference crossing, Jacobi moment and relative-entropy reduction, retained in the scientific collection.

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